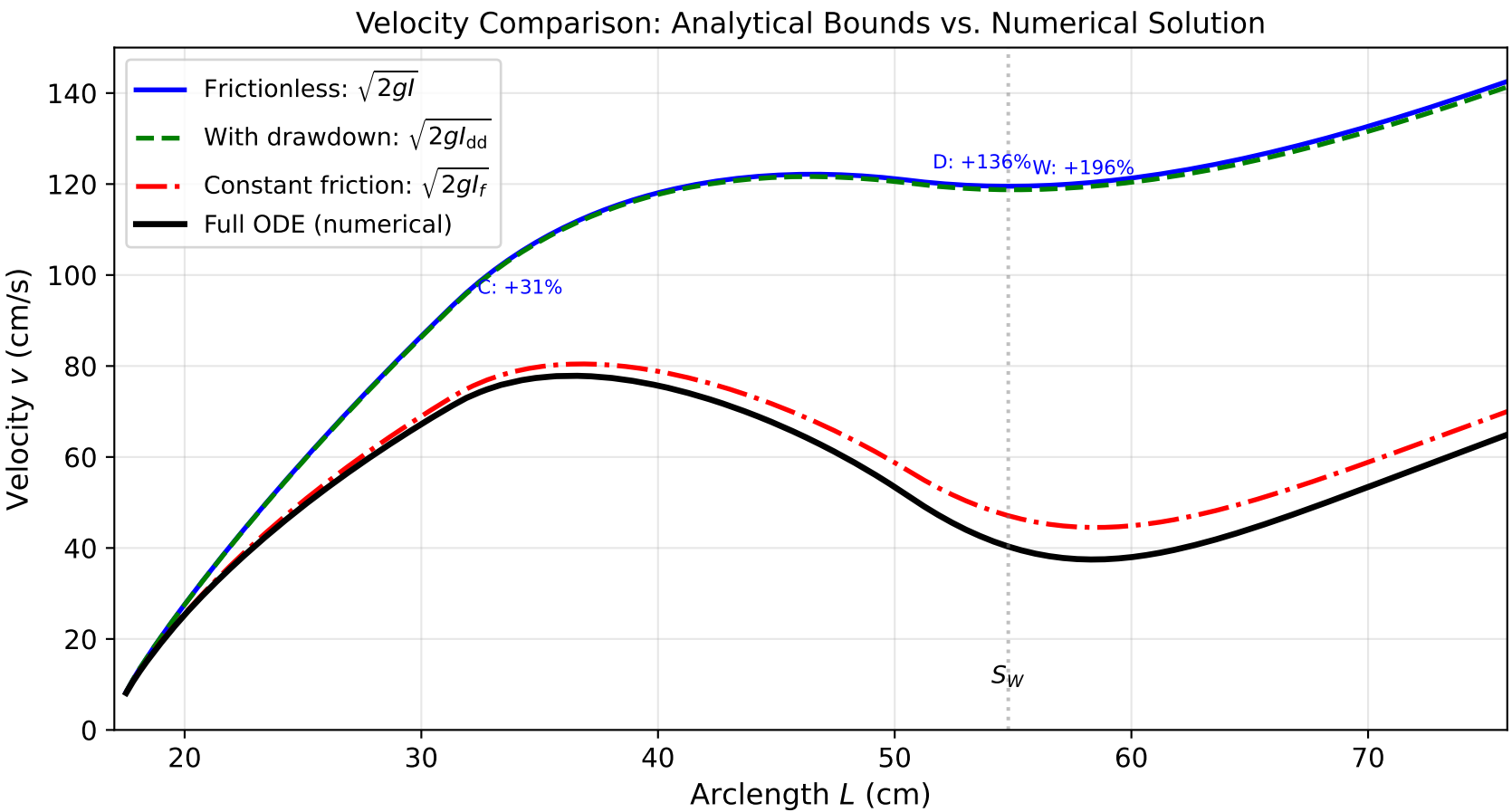
Velocity Comparison: Analytical Bounds vs. Numerical Solution
Frictionless: $\sqrt{2gl}$
With drawdown: $\sqrt{2gl_{dd}}$
Constant friction: $\sqrt{2gl_f}$
Full ODE (numerical)
C: +31%
D: +136%
W: +196%
s_W
Velocity v (cm/s)
Arclength L (cm)
0
20
40
60
80
100
120
140
20
30
40
50
60
70